

Lab 2

JooHo Oh

01-24-2023

Submission Instructions

This lab (and all labs) will be graded for correctness. To submit the lab:

1. Once all code is completed and saved, in the RStudio ribbon, click Knit > Knit to PDF.
2. If you have trouble knitting **anything** to PDF, please work with the Instructor or GSI to resolve the issue. It is okay to instead knit to HTML, then convert to PDF using Adobe or other software.
3. If there is any handwritten work for the lab, combine the handwritten work with the knitted PDF into a single PDF file.
4. Submit the PDF to Gradescope, **selecting** pages for questions. *You will lose half a point on the lab if you don't select pages.*

Lab 1 is due to Gradescope by Sunday, January 29th at 11:59PM PT.

Collaboration Policy

Statistics and Data Science is a collaborative activity. Please feel free to work with others on the lab, but you must **write your solutions individually**. If your lab answers are clearly a slightly altered copy of another student's, this is considered cheating.

Sample US Health Insurance Data

Today we will look at some sample US health insurance data from Kaggle. US insurance charges data tend to suffer from various issues that make modeling difficult:

- sparsity
- right-skewed
- very large values

Q1: Read insurance.csv as object named insurance. Print the dimensions of the data, as well as the data's first few rows.

```
### YOUR CODE HERE:
insurance <- read.csv("~/Desktop/insurance.csv")
dim(insurance)
```

```
## [1] 1338    7
```

```
head(insurance)
```

```
##   age    sex    bmi children smoker   region   charges
## 1  19 female 27.900         0    yes southwest 16884.924
## 2  18  male 33.770         1    no  southeast  1725.552
## 3  28  male 33.000         3    no  southeast  4449.462
## 4  33  male 22.705         0    no northwest 21984.471
## 5  32  male 28.880         0    no northwest  3866.855
## 6  31 female 25.740         0    no  southeast  3756.622
```

```
tail(insurance)
```

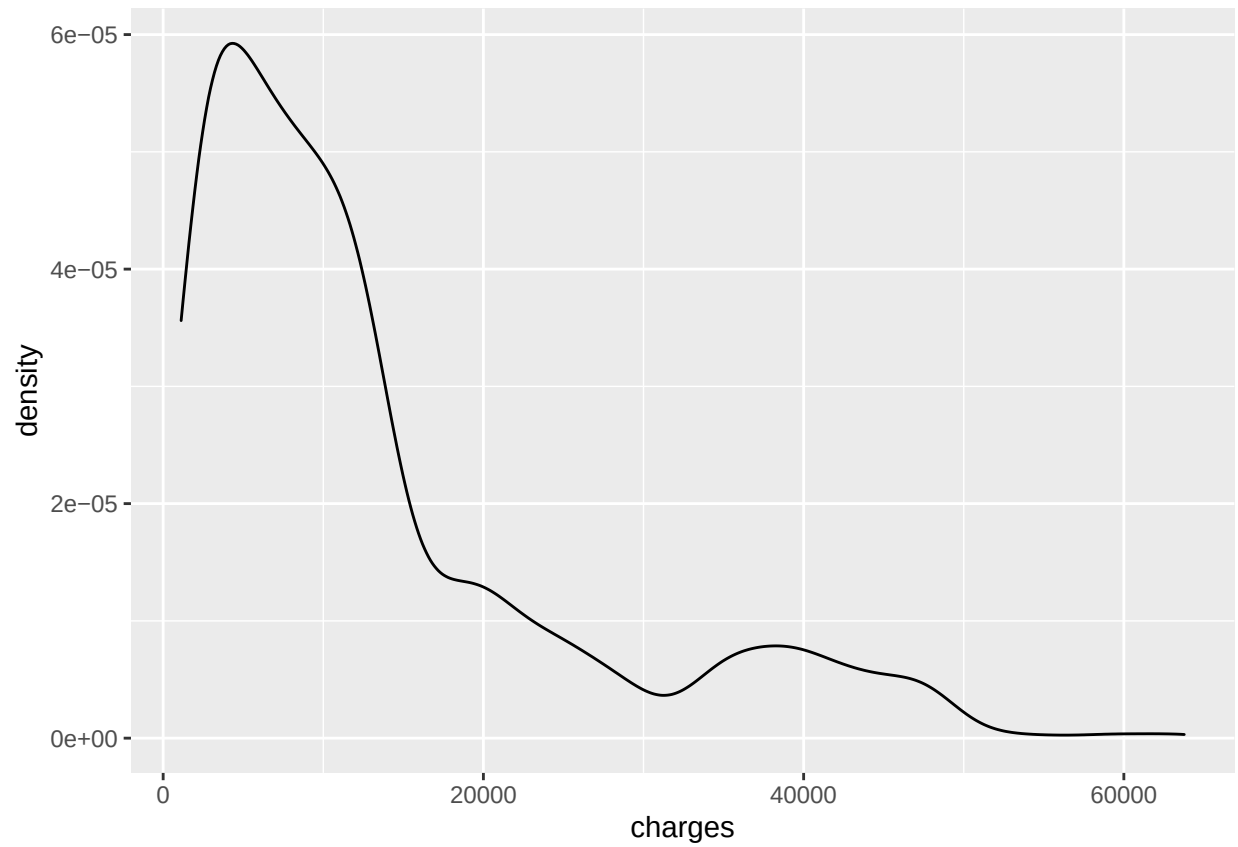
```
##      age    sex    bmi children smoker   region   charges
## 1333  52 female 44.70         3    no southwest 11411.685
## 1334  50  male 30.97         3    no northwest 10600.548
## 1335  18 female 31.92         0    no northeast  2205.981
## 1336  18 female 36.85         0    no  southeast  1629.833
## 1337  21 female 25.80         0    no southwest  2007.945
## 1338  61 female 29.07         0    yes northwest 29141.360
```

The `charges` column are the insurance charges billed to the subscriber.

Q2: Plot the smoothed density curve of the `charges` column, as well as a histogram. Is this variable symmetric, right skewed, or left skewed?

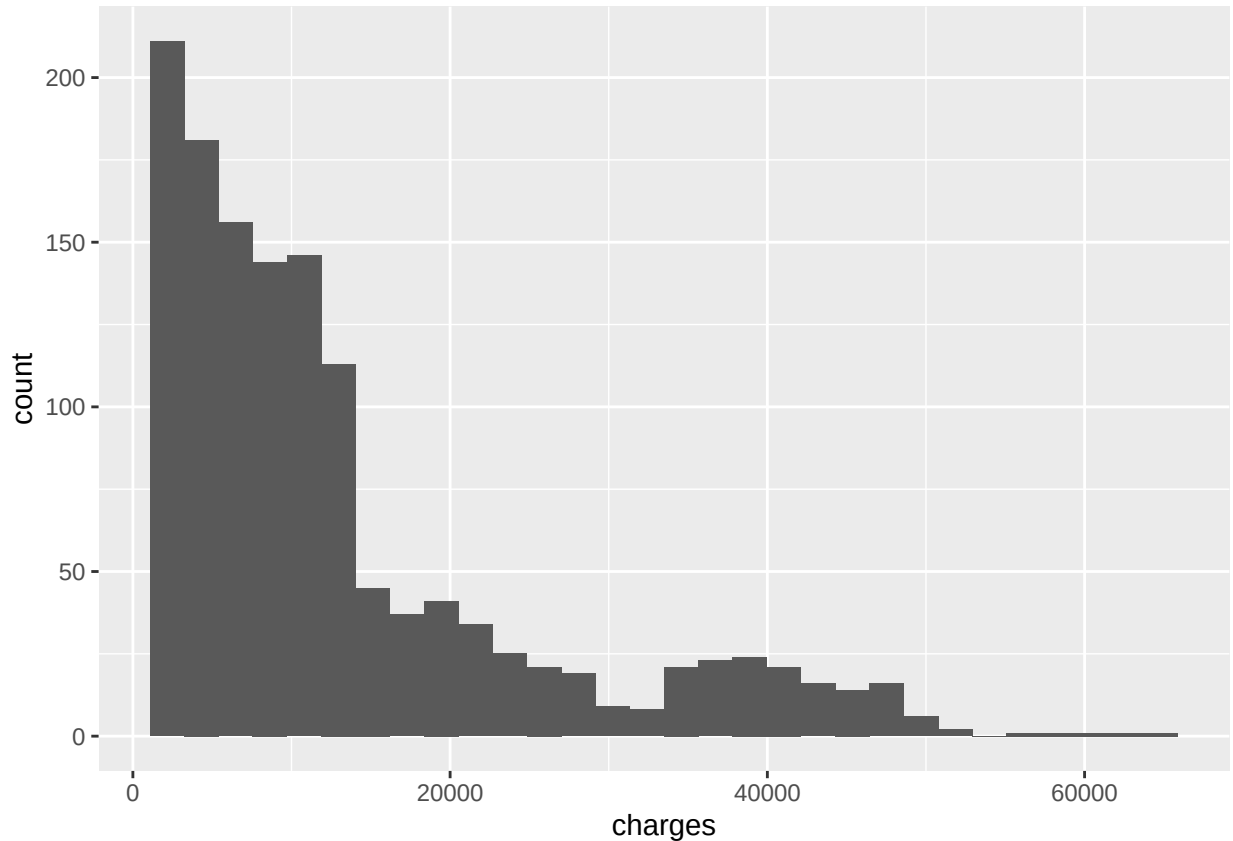
```
### YOUR CODE HERE
```

```
ggplot(data = insurance, aes(x = charges)) + geom_density()
```



```
ggplot(data = insurance, aes(x = charges)) + geom_histogram()
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



YOUR ANSWER HERE: Right-Skewed

Q3: Using code, confirm that there is not sparsity or missingness in the **charges** column.

```
### YOUR CODE HERE
sum(insurance$charges == 0) #is not sparsity

## [1] 0

sum(is.na(insurance$charges)) #no missing values

## [1] 0
```

Luckily, this variable includes **premiums** (ie, monthly insurance cost regardless of whether or not the individual seeks care), so there are not zeroes present in the data.

Q4: Brainstorm some reasons why having a large amount of zeroes (sparsity) in a nonnegative outcome variable might cause problems for linear regression (or symmetrizing transformations)?

Some transformations require you take the log, so the data with 0 does not work for that. There will be a lot of data concentration around zeros, causing our transformation to not be symmetric. Linear regression

coefficients will not be indicative of true model. It will be poor when predicting lower values of our response variable.

Q5: Write a function to compute the “s” statistic (from Lecture 2) for a vector input. Then compute this statistic for the charges variable.

```
### YOUR CODE HERE
computeS <- function(x) {
  s <- (quantile(x, 0.75, names=FALSE) - median(x))/(median(x) - quantile(x, 0.25, names=FALSE))
  return(s)
}

computeS(insurance$charges)

## [1] 1.56361
```

Q6: Write a function to manually compute the BoxCox transformation for a vector input. Then print the “s” statistic for the charges variable for BoxCox(p=2), BoxCox(p=0.5), BoxCox(p=0), BoxCox(p=-1).

```
### YOUR CODE HERE:
boxCox <- function(x, p) {
  if(p==0) {
    return(log(x))
  }
  return((x^p-1)/p)
}

computeS(boxCox(insurance$charges,2))

## [1] 2.881139

computeS(boxCox(insurance$charges,0.5))

## [1] 1.147216

computeS(boxCox(insurance$charges,0))

## [1] 0.8393247

computeS(boxCox(insurance$charges,-1))

## [1] 0.4454304
```

As we would expect, the “s” statistic is increasing as a function of p .

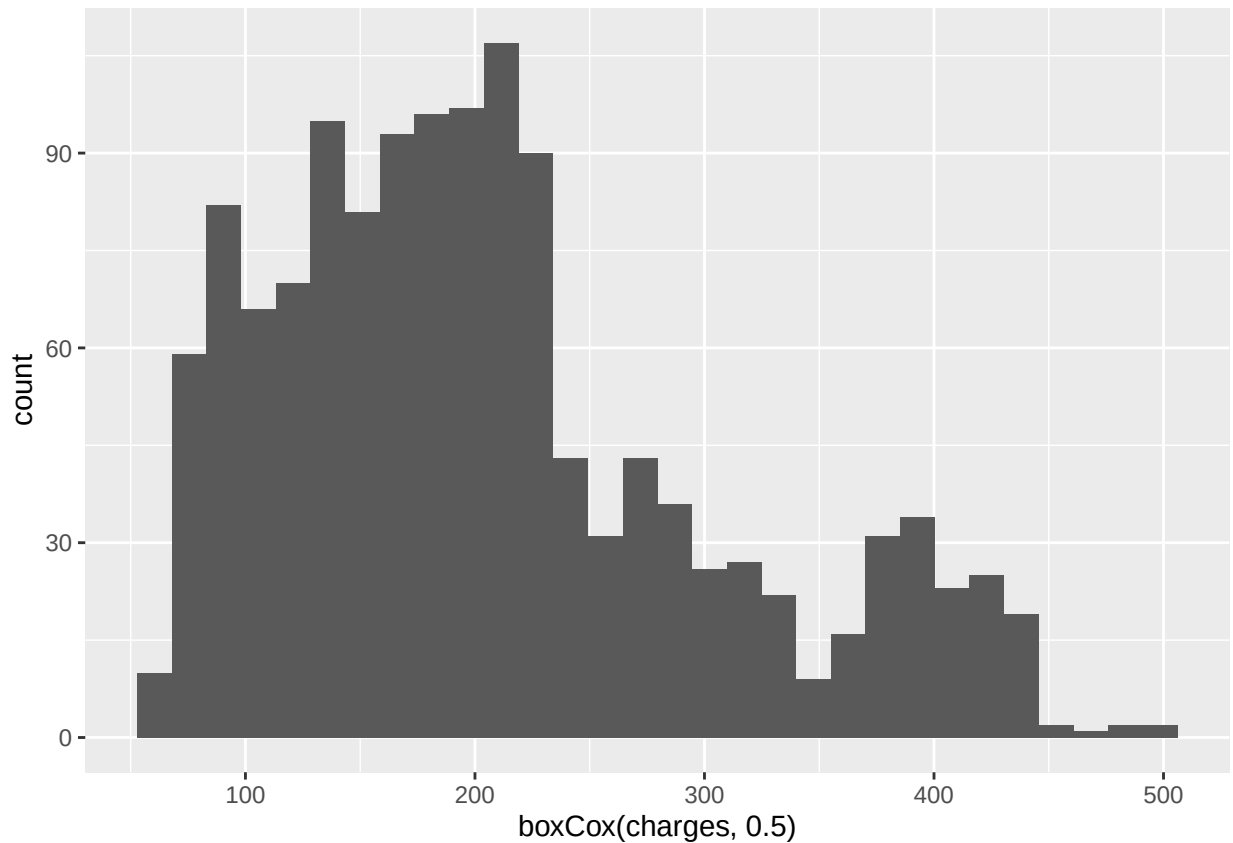
Of the transformations we considered in Q6, $p = 1/2$ and $p = 0$ gives very similar “s” statistic differences to 1.

Q7: Plot two histograms, one for $\text{BoxCox}(p=0)$ and $\text{BoxCox}(p=0.5)$ transformations of charges. Do the two distributions have large differences in symmetry? What does this say about the “s” statistic?

```
### YOUR CODE HERE:
```

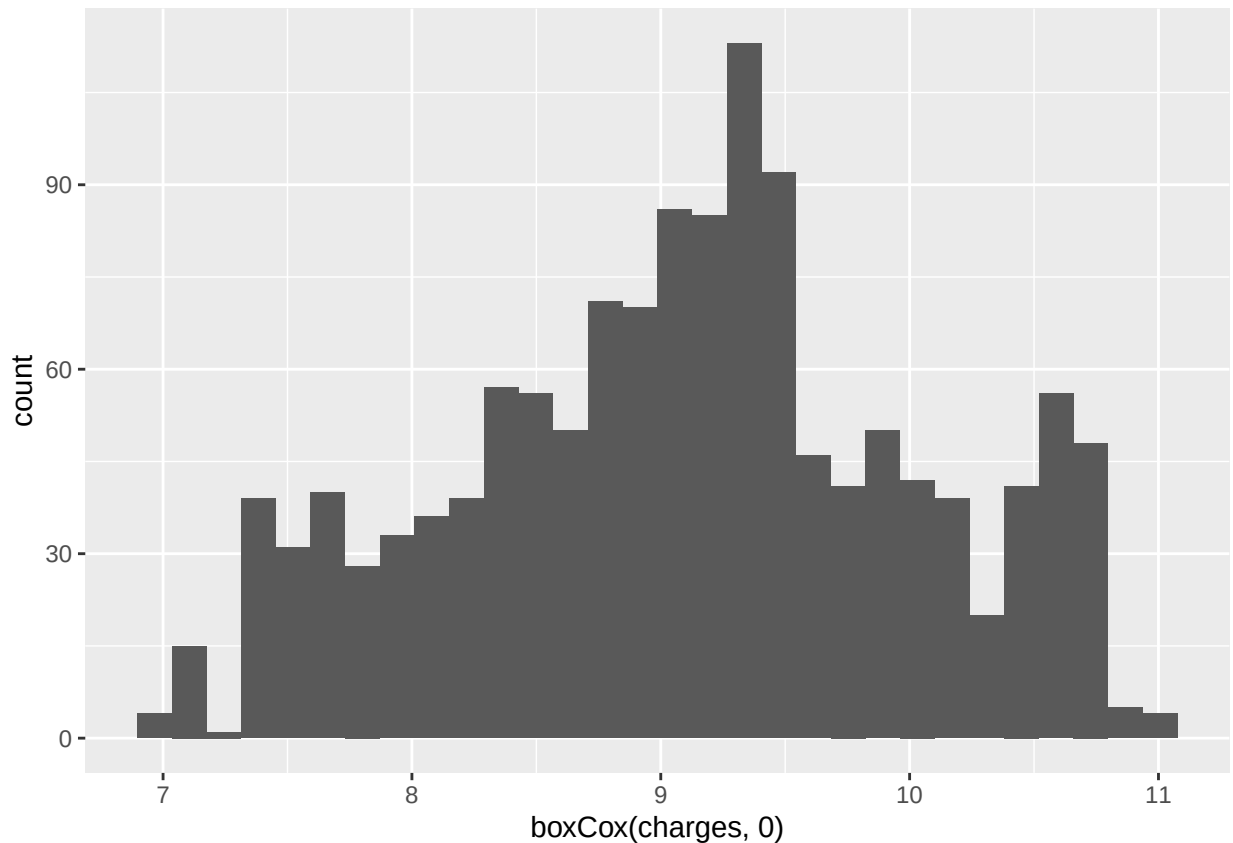
```
ggplot(data=insurance, aes(x=boxCox(charges,0.5))) + geom_histogram()
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



```
ggplot(data=insurance, aes(x=boxCox(charges,0))) + geom_histogram()
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



YOUR ANSWER HERE: $s=0.5$ is more closer to 1 than $s=0$ which means it has to be more symmetric, but in this case if we plot the data it shows us that $s=0$ is more symmetric. So we could say that s statistics does not really apply for all cases.

In general, we should use visualization to sanity check any major preprocessing that we do.

Regression-based selection

Perhaps we can use model-selection to choose the transformation instead. Let's say we are interested in regressing charges on age. Adjusted R-Squared is R-squared (modified for number of covariates we use), and is a measure of the linear model's performance.

Q8: Write a function that takes p is input, and returns the adjusted R-squared of a simple linear model of $\text{boxCox}(\text{charges}, p)$ on age. There is some helper code below to get you started.

```
### YOUR CODE HERE:
modelBoxCoxChargeOnAge <- function(p) {
  insurance_transformed <- insurance %>%
    mutate(charges_transformed = boxCox(charges, p))
  ## TODO: run linear model
  lm_res <- lm(charges_transformed ~ age, data=insurance_transformed)

  sum_res <- summary(lm_res)
```

```

    ## TODO: find and return adj.r.squared in sum_res.
    return(sum_res$adj.r.squared)
}

```

Now that we have this functional model, why don't we try out a bunch of transformations in the range (-2, 2):

```
pseq <- seq(-2, 2, 0.1)
```

Q9: Create a data frame with two columns, one for the p setting in pseq, and one column with the corresponding adjusted R-squared value for the model run with that p setting. Then create a bar plot showing the R-squared results across pseq.

```

### YOUR CODE HERE:
r_sq = rep(0, length(pseq))

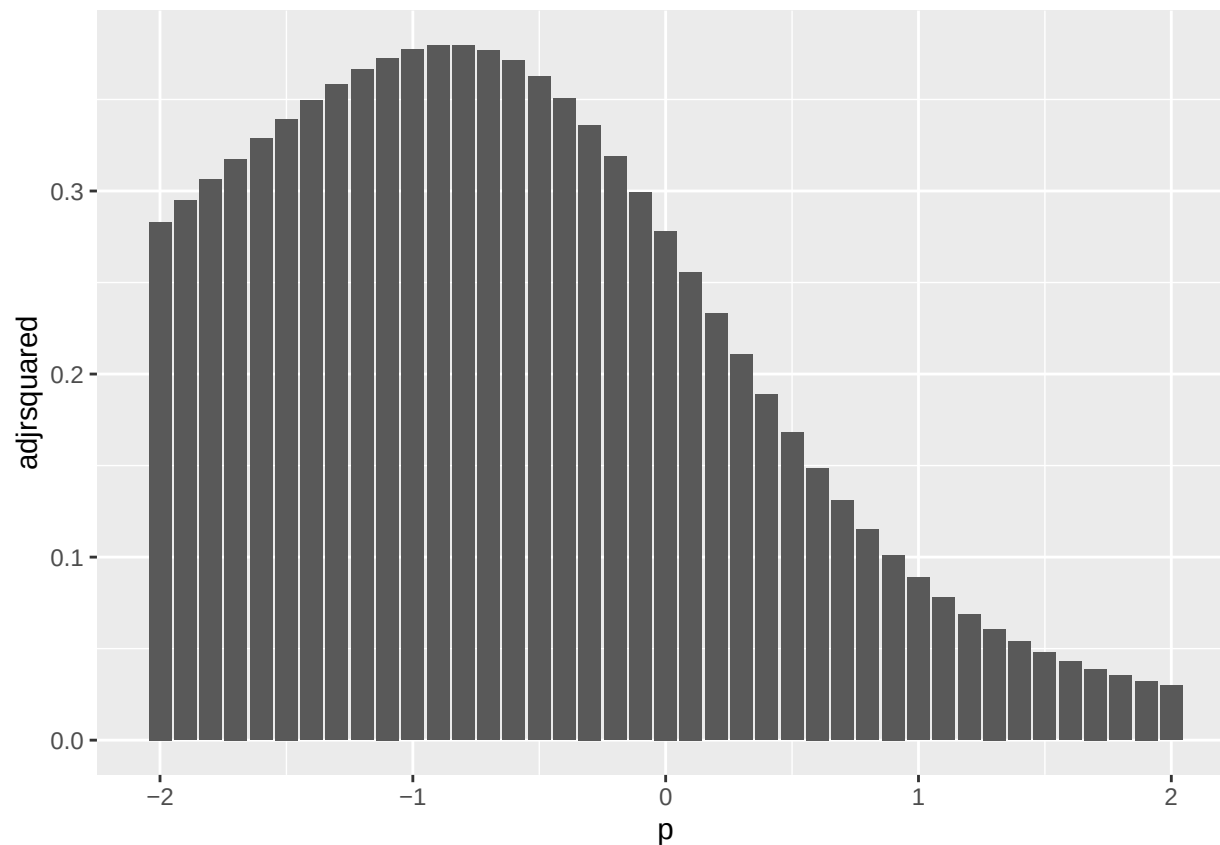
for (i in 1:length(pseq)){
  r_seq = modelBoxCoxChargeOnAge(pseq[i])
}

adj_r_squared = sapply(pseq, modelBoxCoxChargeOnAge)

new_df=data.frame(adjrsquared = adj_r_squared, p=pseq)

ggplot(data=new_df, aes(x=p, y=adjrsquared)) + geom_bar(stat="Identity")

```

We see that by adjusted R-squared criterion, -0.9 is the best BoxCox transformation. This was a lot of settings we tested.

Q10: Any issues you can think of with this approach? Answers are flexible as we will be discussing this topic in more detail later.

YOUR ANSWER HERE: Too complicated of a model would be prone to overfit. Adjust R-squared is not only performance metric, there should be other considerations.

You finished!